

Constraining the multiplicity of Y dwarfs down to planetary masses at 0.5-1000 au with JWST

Institut Trottier de recherche sur les exoplanètes
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Université de Montréal

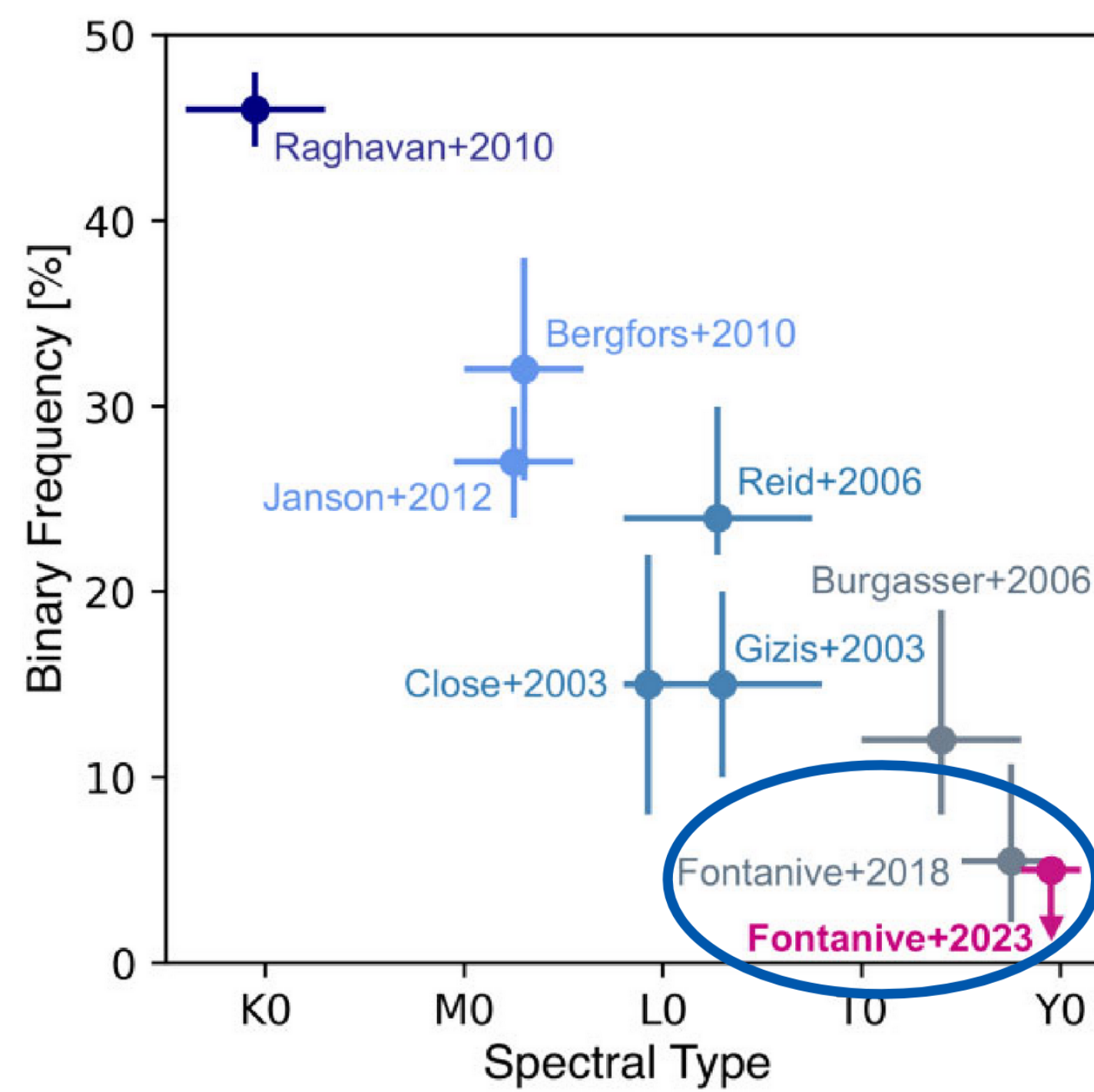
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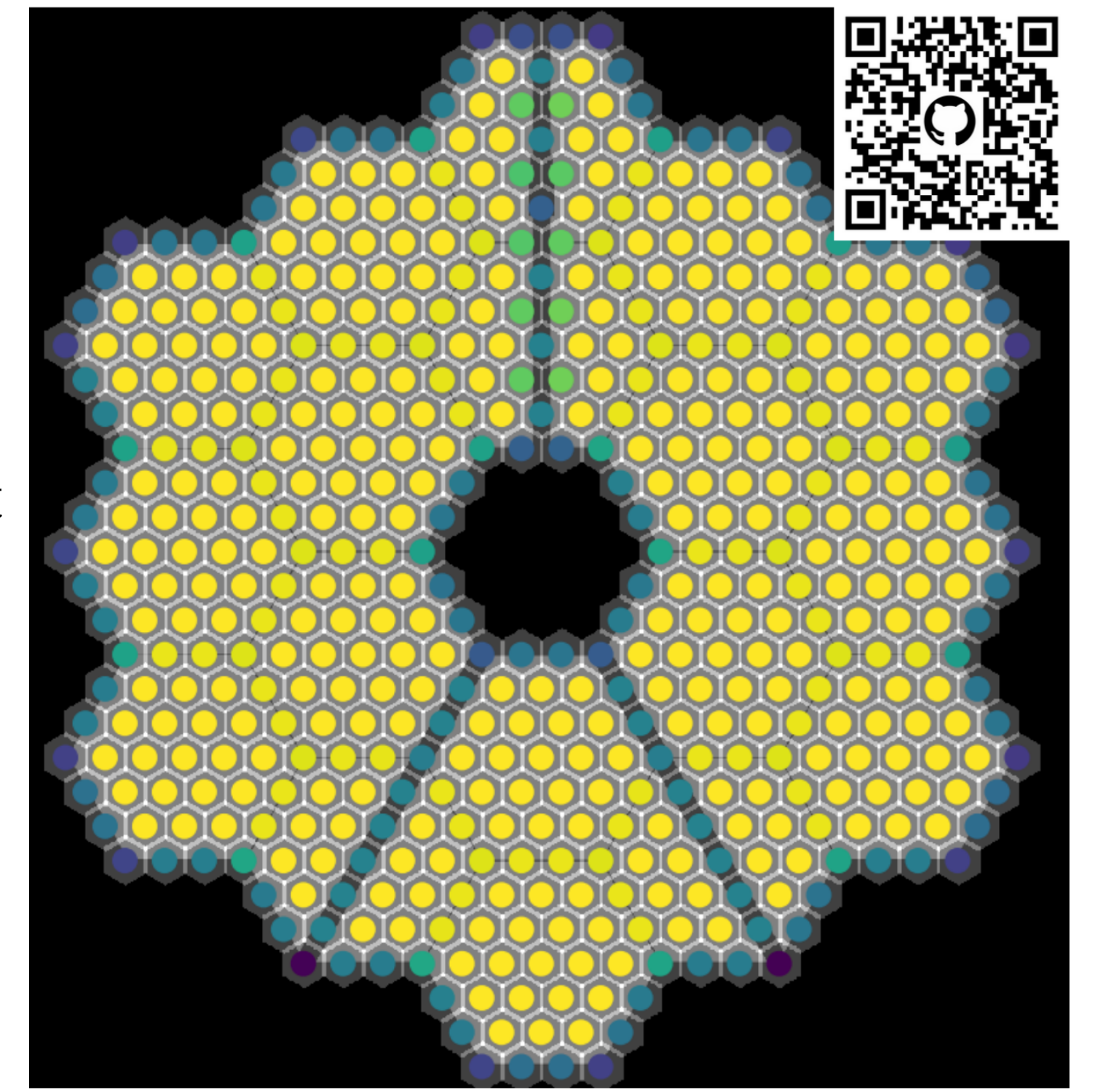
Multiplicity survey of 22 Y dwarfs with JWST



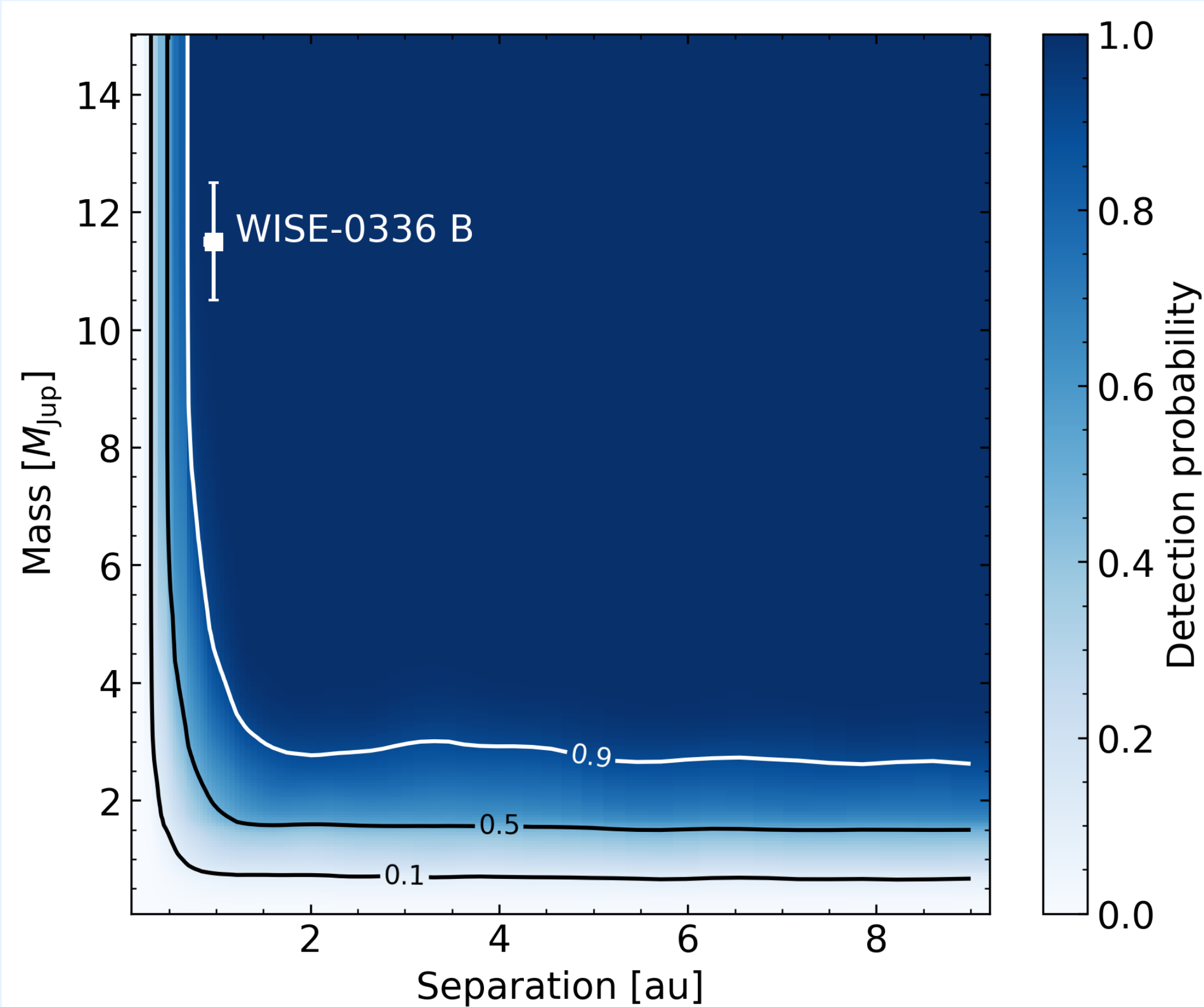
- Multiplicity can be used to distinguish stellar formation mechanisms that produce different binary fractions
- Multiplicity decreases throughout the brown dwarf sequence
- However, for Y dwarfs **HST results were limited to a few objects** and separations beyond 2 AU [1, 2]
- To confirm this trend and probe short separations, **we surveyed 22 Y dwarfs with JWST NIRCам and NIRISS at 4.8 μm**

Kernel Phase: imaging below the diffraction limit

- We analyze the JWST images with Kernel Phase Imaging: we **model the telescope as an interferometer** made of small sub-apertures [3]
- This unlocks relatively high contrasts below the diffraction limit of JWST
- The pipeline is available on GitHub!
- We **recover the Y+Y dwarf binary WISE-0336 AB** with this technique and obtain parameters consistent with ePSF fitting [4]



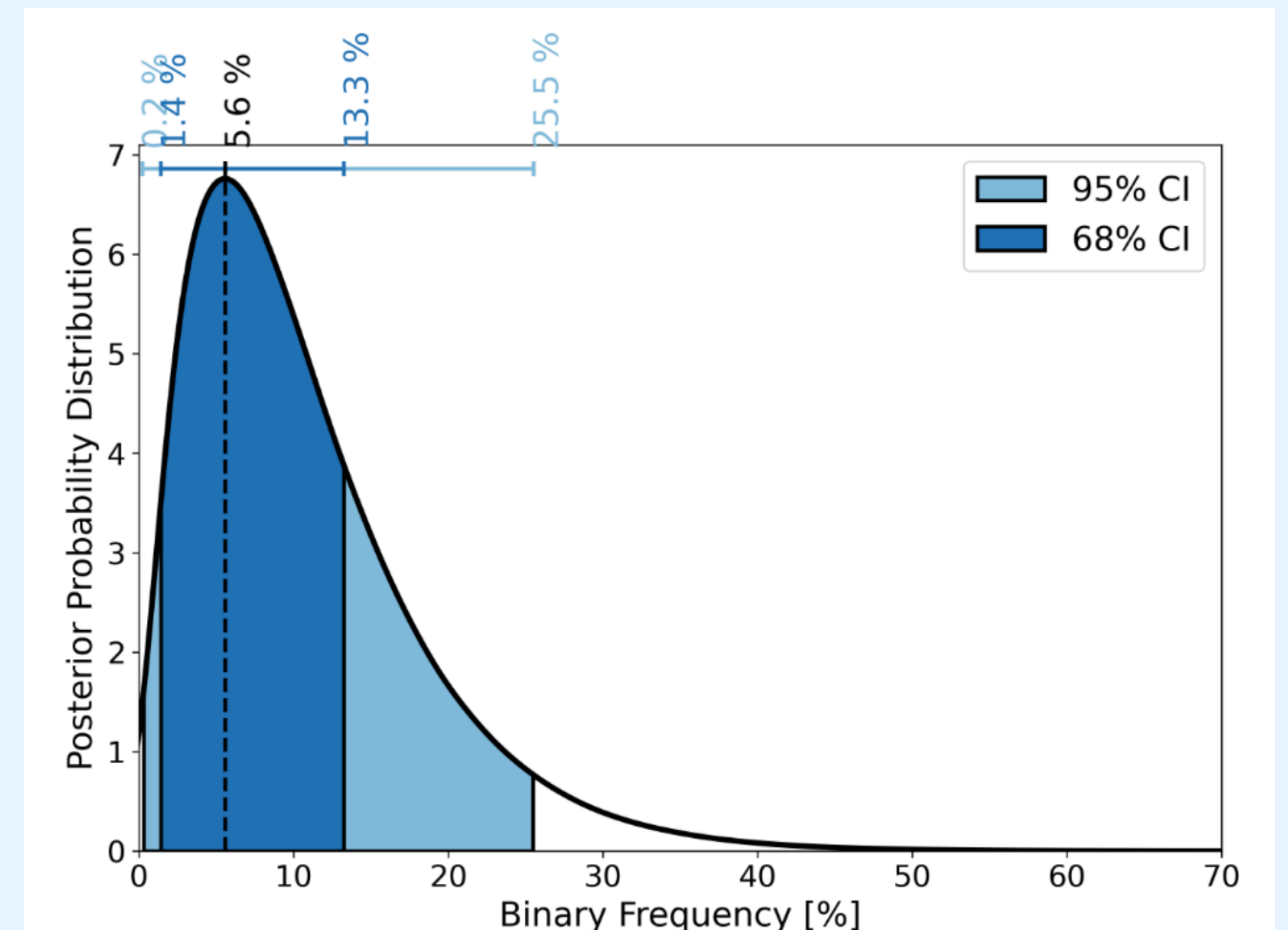
We are sensitive to companions down to 3 M_{Jup} below 2 AU with 90 % completeness and constrain to Y dwarf multiplicity to $5.6^{+7.7}_{-4.2}$ %



We use our kernel phase contrast limits to derive completeness map for the survey (left)

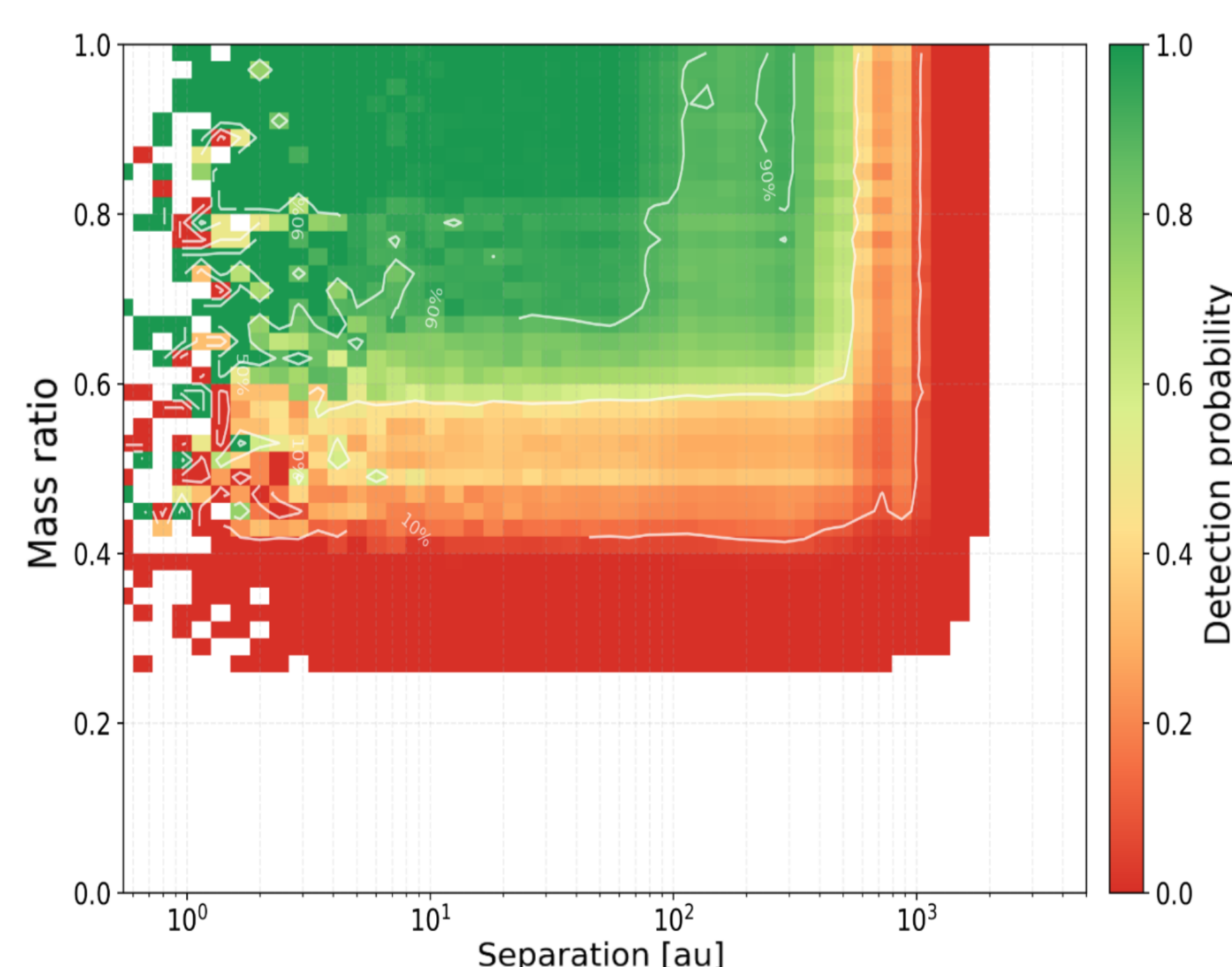
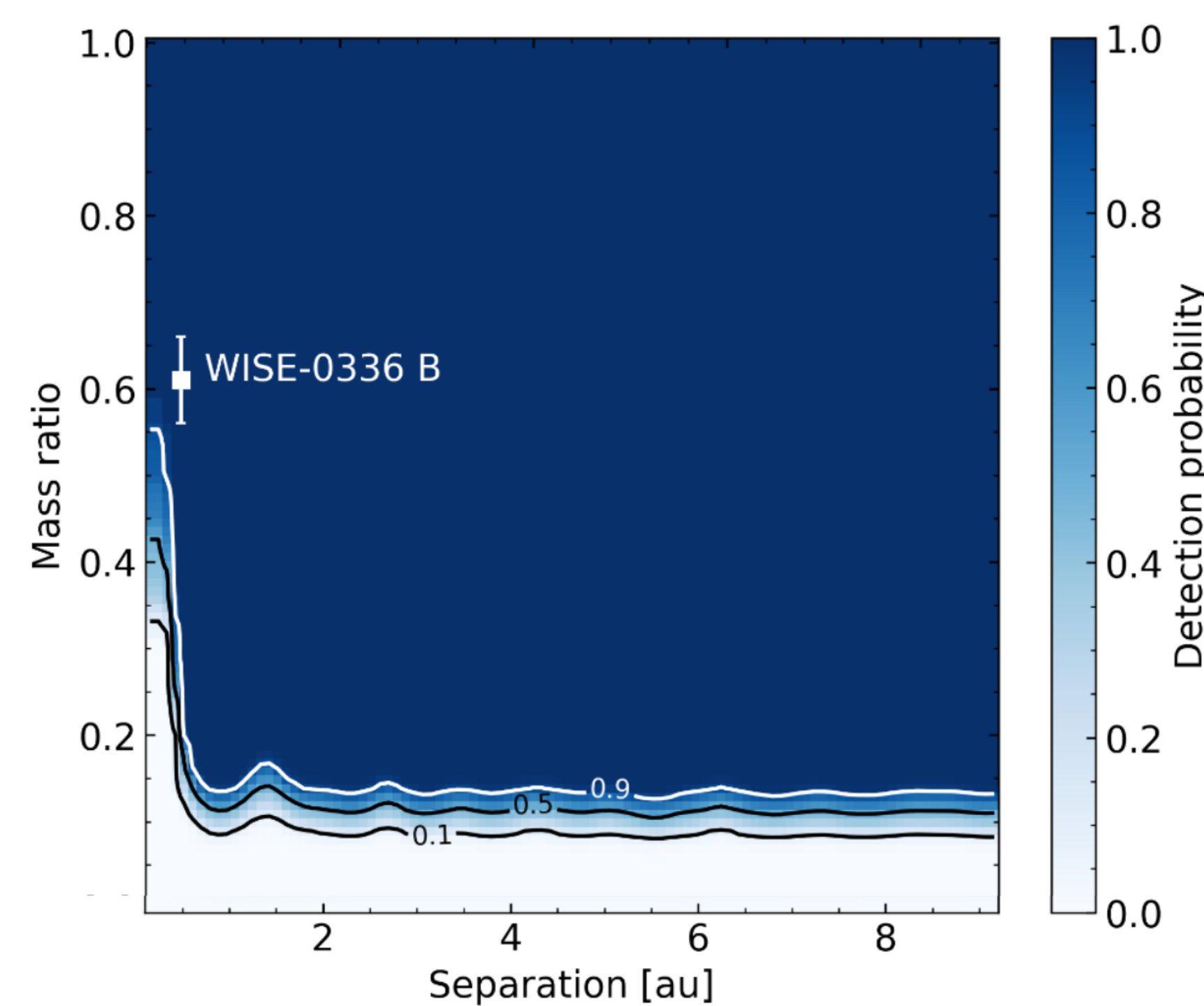


We then use this completeness map and synthetic populations to estimate the binary fraction in a Bayesian framework (right)



We also performed a wide separation search up to 1000 au

No wide companions were found. We are currently combining the completeness maps in the two regimes to derive a global completeness and binary fraction.



Future Work

- Papers coming up: Vandal et al. (in prep), Giovinnazzi et al. (in prep)
- Detailed comparison between kernel phase and ePSF fitting
- JWST Cycle 5 Survey 10110 (P.I. Fontanive, Co-P.I. Vandal) to search for companions around more of the coolest brown dwarfs
- Follow-up WISE-0336: Track the orbit to get dynamical mass measurements

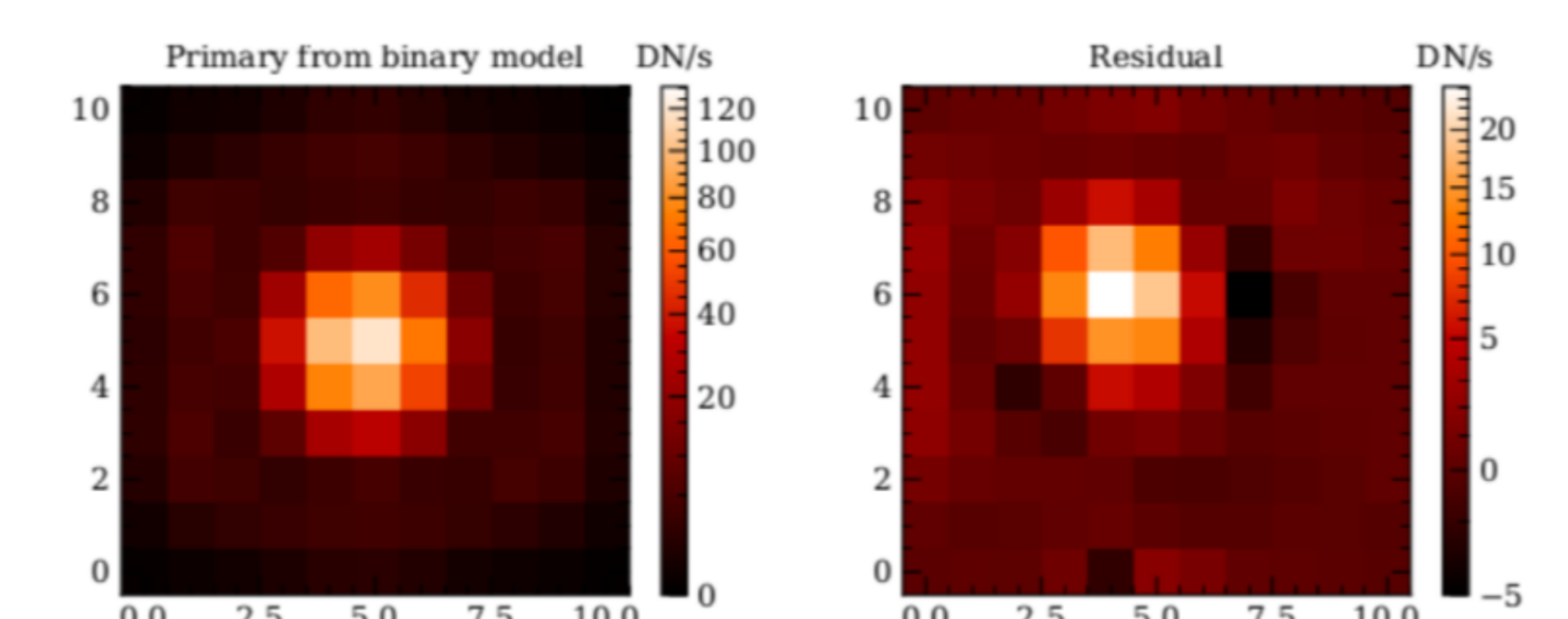


Figure showing the ePSF discovery of WISE-0336 B [4]

References

- Fontanive et al., 2018, *MNRAS*, 479, pp. 2702–2727.
- Fontanive et al., 2023, *MNRAS*, 526, pp. 1783–1798.
- Martinache, 2010, *ApJ*, 724, pp. 464–469.
- Calissendorff et al., 2023, *ApJ*, 947, p. L30.

See the PDF version of this poster!

